



# Air Quality Situation Report

Edmonton Symphony Orchestra Outdoor Concert · Hawrelak Park (Heritage Amphitheatre)

Thu, Sep 3, 2026 · 1:07 AM MDT

## Air Quality Health Index (AQHI) — Edmonton

# 2.0

Current AQHI at Edmonton Lendrum (3.51 km from venue)  
Forecast Thu 2:00 AM at Edmonton Lendrum (3.51 km from venue): **2.1**

| Health Risk | AQHI | At-Risk Population                   | General Population                        |
|-------------|------|--------------------------------------|-------------------------------------------|
| Low Risk    | 2.1  | Enjoy your usual outdoor activities. | Ideal air quality for outdoor activities. |

No active Environment Canada weather alerts for the venue.

## Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 28.4°C and feels near 29.1°C. Winds are 18 km/h from SW, gusting near 33 km/h.
- The nearest current AQHI is 2 at Edmonton Lendrum, 3.5 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 2 (forecast tonight: 3, tomorrow: 3).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 4.0 km away) puts the area near 1.
- The nearest air monitoring station (Edmonton Lendrum, 3.5 km) reports fine particulate matter (PM2.5) at 5.0 µg/m³.
- A nearby community sensor ('ACA\_LaurierHeightsCL', 1.8 km) reads 2.6 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 152.4 km to the SE, i.e. arriving from the SE.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-09-02) is 10.2 km E of the venue. The modeled back-trajectory passes within 6.0 km of this fire, making smoke transport to the venue plausible based on wind-field modeling.
- The AQHI forecast for Thu 2:00 AM is 2, based on Edmonton Lendrum, 3.5 km from the venue.
- Thunderstorm conditions appear in the hourly forecast beginning around Thu 6:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are heat, precipitation, thunderstorm.

## Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Increase hydration, shade, cooling and heat-illness messaging.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

## Hazard Snapshot — Overall Risk: High Risk

|                                                                |                                                      |                                                   |                                                                      |                                                                       |
|----------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------|
| AIR QUALITY<br><b>Low Risk</b><br>2.1 AQHI                     | HEAT<br><b>High Risk</b><br>33.2 °C apparent/humidex | WIND<br><b>Moderate Risk</b><br>64 km/h peak gust | PRECIPITATION<br><b>High Risk</b><br>5.2 mm/h maximum                | THUNDERSTORM<br><b>High Risk</b><br>Thu 6:00 PM first forecast signal |
| AQHI RATE OF CHANGE<br><b>Low Risk</b><br>0.0 AQHI change/hour |                                                      |                                                   | WIND SHEAR<br><b>Low Risk</b><br>21.8 ° direction diff (10m vs 120m) |                                                                       |

## Local Monitoring

|                                                             |                                                   |                                                                   |                                                                        |
|-------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|
| Nearest station<br><b>2.0</b><br>Edmonton Lendrum · 3.51 km | Blended estimate<br><b>1.0</b><br>confidence High | PM2.5 (station)<br><b>5.0 µg/m³</b><br>Edmonton Lendrum · 3.51 km | PM2.5 (community)<br><b>2.6 µg/m³</b><br>ACA_LaurierHeightsCL · 1.8 km |
|-------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|

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Prepared by Alberta Capital Airshed for Edmonton Symphony Orchestra outdoor concerts. Beta decision-support product  
— for the interactive map, live cameras and hourly outlook, scan the QR code or visit [https://dkevinm.github.io/](https://dkevinm.github.io/eso_sitre/)

